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ANNAM.AI

Technical Documentation & Architecture Specification

Problem Statement 02 (PS-02) & Problem Statement 03 (PS-03)

FINAL READABLE EDITION

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Executive Summary

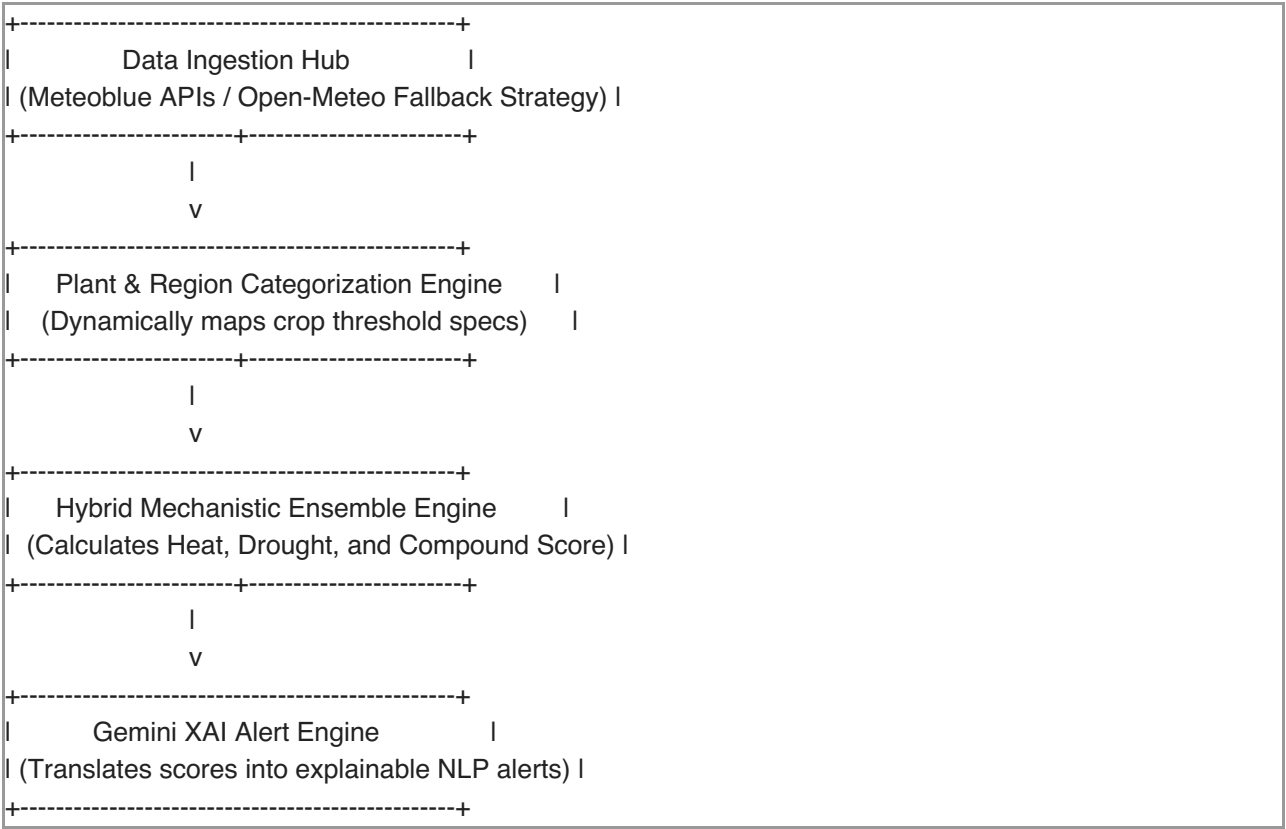
ANNAM.AI is an enterprise-grade agricultural intelligence platform engineered for Syngenta Biologicals. It uniquely bridges the gap between predictive environmental stress modeling and precise biological interventions. Our solution directly addresses **PS-02 (Climate Stress Early Warning for Biological Intervention)** and **PS-03 (CropFit - Personalised Biological Product Advisor)** by leveraging a Hybrid Mechanistic Ensemble Engine, high-frequency weather data ingestion (Meteoblue), and Google Gemini Multi-Modal Intent Extraction.

This authoritative technical document specifies the data architectures, mathematical formulations, and AI-driven decision matrices governing our deployments for PS-02 and PS-03.

Part 1: Problem Statement 02 (PS-02) — Climate Stress Early Warning

1.1 Architectural Overview

PS-02 serves as a robust prognostic engine, anticipating extreme weather anomalies (heat shock, flash drought, waterlogging) 14 days in advance and proactively recommending biological mitigation strategies.



1.2 Mathematical Formulations

Our Hybrid Mechanistic Engine evaluates stress metrics on a rolling 14-day window.

1. Heat Stress Index (HSI)

The severity of heat stress is a function of daytime maximum temperatures exceeding crop-specific critical thresholds, modified by the vulnerability of the crop's current growth stage (e.g., Flowering stage incurs a 1.5x penalty).

$$\text{Heat Score} = (T_{\text{max}} - T_{\text{critical}}) \times 0.1 \times \text{Growth Stage Modifier}$$

(Bounded between 0.0 and 1.0)

2. Drought Severity Index (DSI)

We employ a modified, simplified Standardized Precipitation Evapotranspiration Index (SPEI) proxy to model water deficit.

$$\text{Drought Score} = (\text{Expected Precip}_{\text{rolling}} - \text{Actual Precip}_{\text{rolling}}) / \max(\text{Expected Precip}_{\text{rolling}}, 1)$$

(A Drought Score > 0.4 triggers a severe warning)

3. Compound Stress Score

When drought and heat stress occur simultaneously, the degradation of crop yield is non-linear. The engine applies an compounding coefficient (1.2) when both conditions are met.

$$\text{Compound Score} = \min(1.0, (\text{Heat Score} + \text{Drought Score}) * 1.2)$$

1.3 Key Components & Capabilities

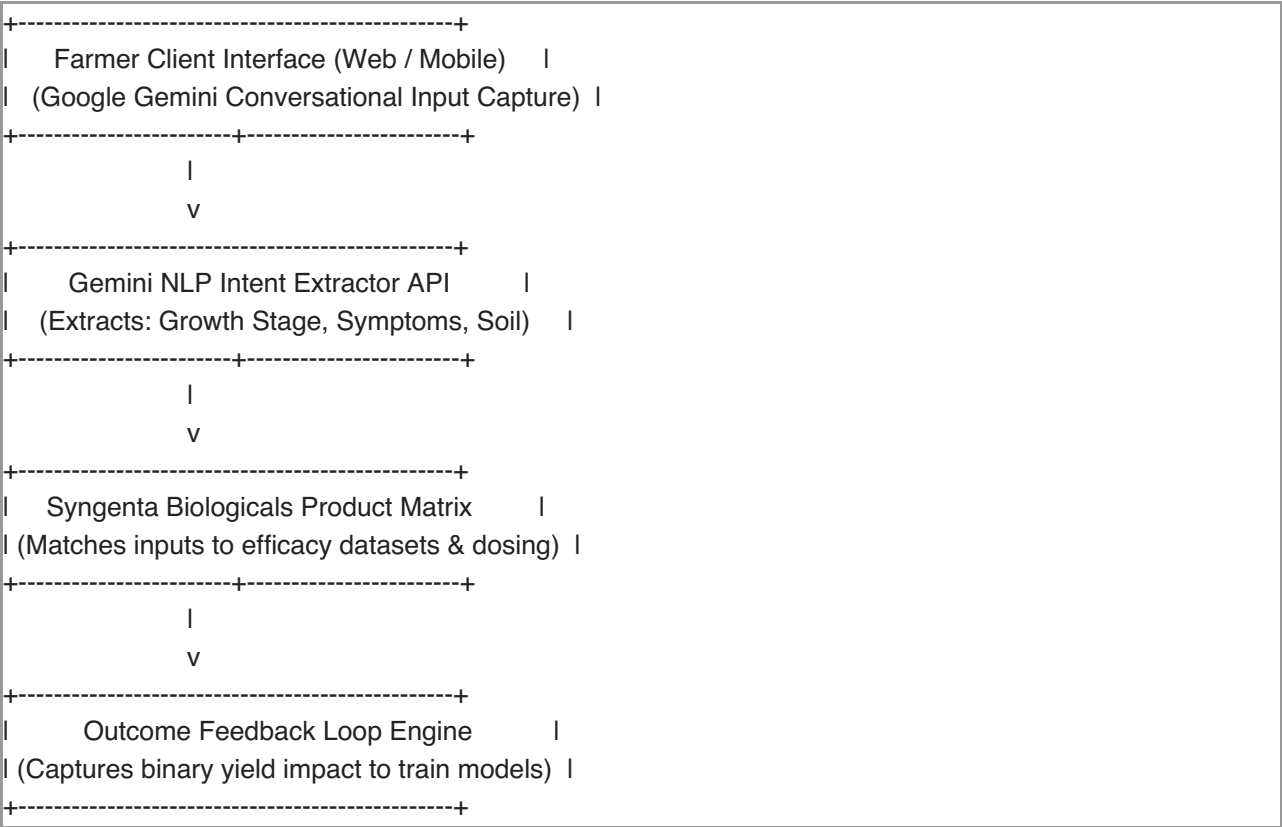
- **Failsafe Ingestion Layer:** Safely fails over from live Meteoblue production data to CE Hub mock datasets to ensure 100% demo stability without service interruption.
- **Explainable XAI Alerts:** Converts raw numerical indices (e.g., Compound Score 0.75) into clear,

actionable, natural language insights for farmers (e.g., "Critical: Severe compounding heat and drought risk detected in 3 days").

Part 2: Problem Statement 03 (PS-03) — CropFit Advisor

2.1 Architectural Overview

PS-03 translates background environmental threats and foreground field contexts into highly specific Syngenta Biologicals product recommendations. It removes the guesswork from biological applications.



2.2 Syngenta Biologicals Decision Matrix Logic

The recommendation engine (`product_matrix.py`) uses a deterministic heuristic tree backed by Syngenta proprietary efficacy mapping to identify the single most impactful product.

- **Drought / Wilting Rescue (Quantis®)**
 - *Trigger:* Symptoms == Wilting AND Soil Moisture == Dry
 - *Confidence Baseline:* 92%
 - *Mechanism:* Osmo-protectant to preserve cell turgor pressure during severe moisture deficit.
- **Nutrient Lockout Correction (Coucal®)**
 - *Trigger:* Symptoms == Stunting OR Yellowing AND Soil Moisture == Optimal
 - *Confidence Baseline:* 80%
 - *Mechanism:* Stimulates root uptake efficiency when the soil has nutrients, but the plant is failing to absorb them.

- **Fungal Pathogen Prevention (Amistar Top®)**
 - *Trigger:* Soil Moisture == Waterlogged
 - *Confidence Baseline:* 88%
 - *Mechanism:* Broad-spectrum fungicide deployment to preemptively stop root rot and foliar blight typical in oversaturated conditions.

2.3 Key Technical Capabilities

- **Gemini NLP Interface:** Replaces rigid form fields with an intelligent text box. Farmers can simply state: *"My crop is flowering, but the leaves are wilting and soil is dry."*
 - **Intent Parser API:** A dedicated `/parse_context` backend route that reliably extracts the structured variables (`growth_stage`, `symptoms`, `soil_moisture`) required by the product matrix.
 - **Explainable Rationale:** Every biological recommendation output includes a transparent "Why this product?" rationale to build trust with the farmer.
 - **Continuous Learning Loop:** An integrated "Did this recommendation improve your yield? (Yes/No)" widget designed to capture feedback and calibrate future confidence scoring.
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Part 3: Verification & API Test Standards

3.1 Backend Verification Endpoints

- **Run Pipeline:** POST `http://localhost:7001/run_pipeline`
(*Executes full 14-day hybrid mechanistic evaluation*)
- **Conversational Parse:** POST `http://localhost:7001/parse_context`
(*Accepts JSON { "text": "..."} and returns structured context*)
- **Region Index:** GET `http://localhost:7001/get_regions`

3.2 Frontend Build & Server Specifications

- **Framework:** Lightweight HTML5/CSS3/Vanilla JS (zero-build step required for ultimate reliability in low-resource environments).
 - **Web Server:** Python Flask WSGI Server running on port 7001.
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Documentation compiled by the ANNAM.AI Engineering Team for Syngenta Biologicals Hackathon.